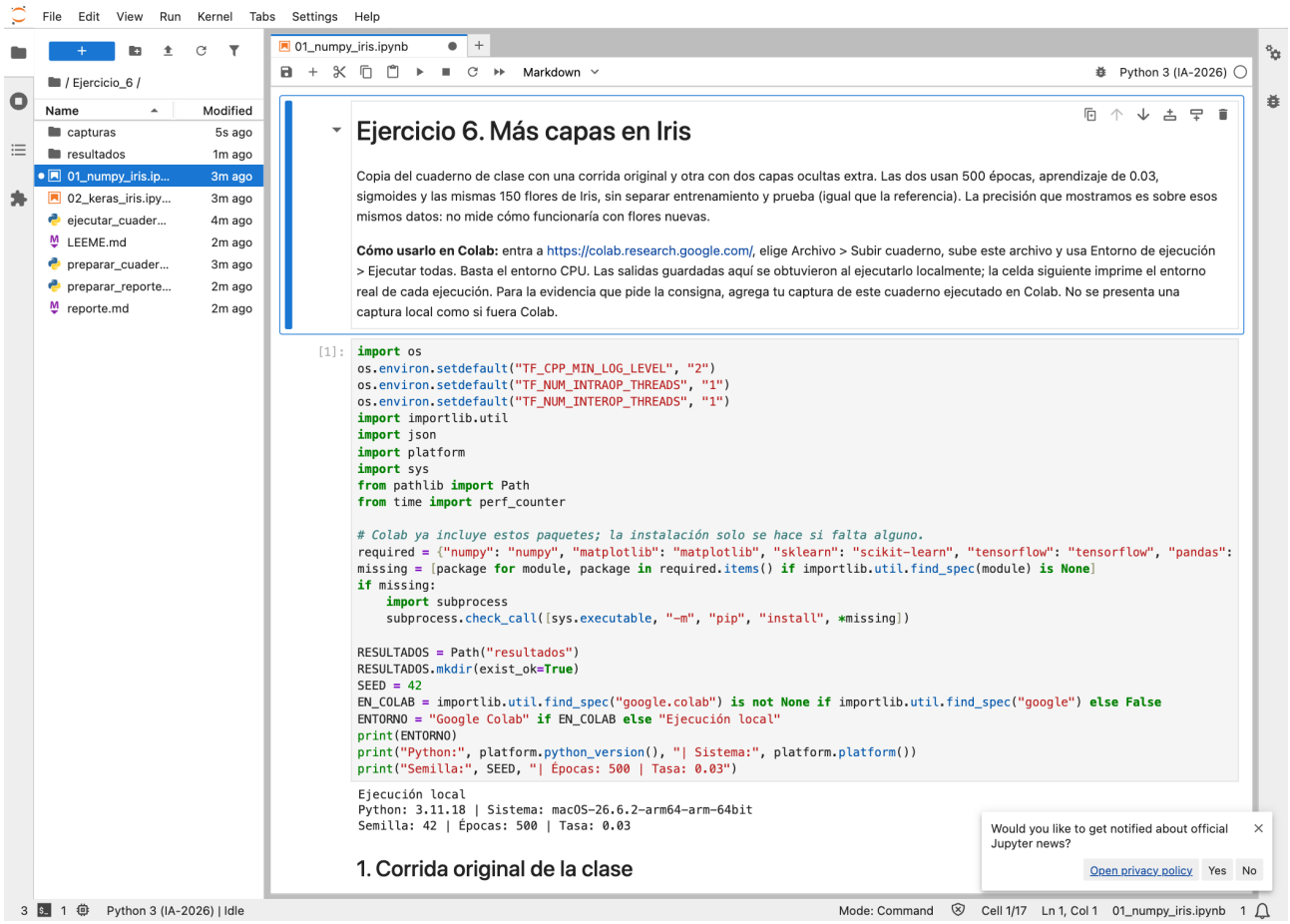
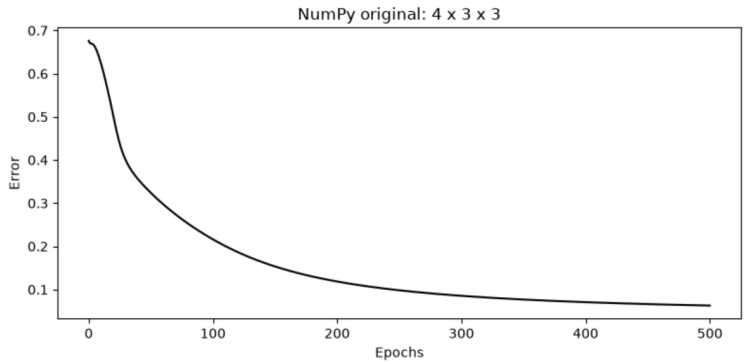
# Ejercicio 6. Capturas

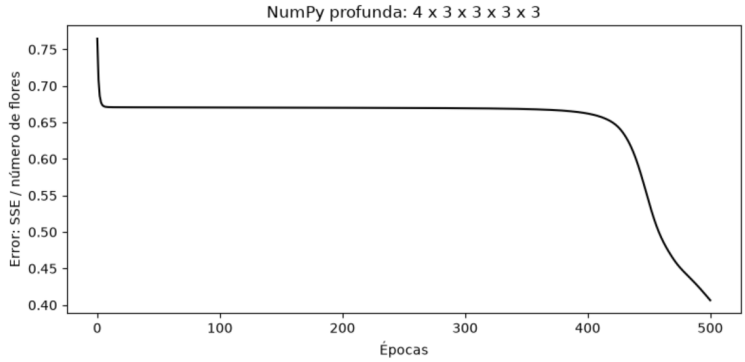



NumPy: red original.



## NumPy: red con dos capas extra.

```
CORRIDA PROFUNDA NUMPY: 4 x 3 x 3 x 3 x 3
Época  50: 0.670597
Época 100: 0.670392
Época 150: 0.670225
Época 200: 0.670054
Época 250: 0.669827
Época 300: 0.669417
Época 350: 0.668285
Época 400: 0.662153
Época 450: 0.540626
Época 500: 0.406424
topologia : [4, 3, 3, 3, 3]
error_sse_por_muestra_final : 0.406423832817962
mse_final : 0.13547461093932067
precision_entrenamiento : 0.7066666666666667
segundos : 2.9070119999996678
Ejemplo [3,3,1,1]: [0.62814511 0.42621388 0.0442126 ]
```




## Keras: resumen de la red original.

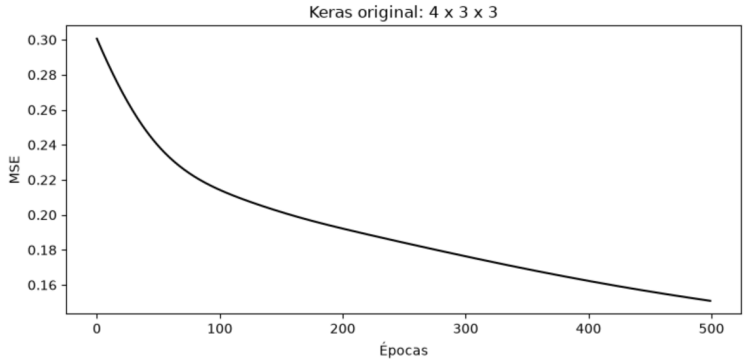
**Model: "sequential"**

| Layer (type) | Output Shape | Param # |
|---|---|---|
| layer1 (Dense) | (None, 3) | 15 |
| layer2 (Dense) | (None, 3) | 12 |

**Total params:** 27 (108.00 B)
**Trainable params:** 27 (108.00 B)
**Non-trainable params:** 0 (0.00 B)

Out[6]: 655

## Keras: curva de la red original.



Keras: resumen y curva de la red con dos capas extra.

**Model: "sequential_1"**

| Layer (type) | Output Shape | Param # |
|---|---|---|
| layer1 (Dense) | (None, 3) | 15 |
| layer2 (Dense) | (None, 3) | 12 |
| layer3 (Dense) | (None, 3) | 12 |
| layer4 (Dense) | (None, 3) | 12 |

**Total params:** 51 (204.00 B)

**Trainable params:** 51 (204.00 B)

**Non-trainable params:** 0 (0.00 B)

```
CORRIDA PROFUNDA KERAS
topologia : [4, 3, 3, 3, 3]
loss_ultima_epoca : 0.22171808779239655
mse_final : 0.22171393036842346
precision_entrenamiento : 0.5
segundos : 5.197332499999902
parametros : 51
Época  50: loss = 0.256202
Época 100: loss = 0.235026
Época 150: loss = 0.227257
Época 200: loss = 0.224189
Época 250: loss = 0.222882
Época 300: loss = 0.222289
Época 350: loss = 0.222004
Época 400: loss = 0.221857
Época 450: loss = 0.221773
Época 500: loss = 0.221718
Predicción [3,3,1,1]: [0.34232172 0.33883744 0.33637056]
```

